# MANEESH KUMAR CHINTAKINDI

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## SUMMARY

AI/ML engineer with an M.S. in Artificial Intelligence, spanning classical machine learning, deep learning, and generative AI - from CNN/NLP classification systems and speech emotion recognition to GenAI applications built with RAG, LangChain, and FAISS. Brings a software engineering foundation from 2+ years automating backup and recovery systems at Ericsson, combined with applied ML/GenAI research through graduate coursework and independent projects. Azure AI Engineer Associate certified; comfortable across PyTorch, TensorFlow, FastAPI, and MLflow-driven experimentation, with a track record of clear technical documentation and cross-functional collaboration.

## EDUCATION

**M.S. in Artificial Intelligence** | University of North Texas, Denton, TX GPA: 3.6 / 4.0 | May 2026

*Coursework: Machine Learning, Deep Learning, NLP, Generative AI, Data Mining, Data Visualization, Software Engineering*

**B.Tech in Computer Engineering - AI & ML Specialization** | Ganpat University, India GPA: 8.32 / 10 | May 2024

## TECHNICAL SKILLS

**GenAI & LLMs:** LLM Fine-tuning (LoRA, PEFT, RLHF), Prompt Engineering, RAG, LangChain, FAISS, Hugging Face Transformers, Generative AI, Structured Output Generation, Function Calling, Chatbot Development, Agentic Workflows

**Software Development:** Python (primary), R, C, JavaScript, SQL, NoSQL, MySQL | FastAPI, Flask, Django, Streamlit, REST APIs, Git, Docker, OOP, SDLC, Agile, Unit Testing, CI/CD concepts

**ML & Data:** PyTorch, TensorFlow, Scikit-learn, XGBoost, CNNs, RNNs, Computer Vision, NLP, Text Classification, Sentiment Analysis, Time Series Analysis, Feature Engineering, Model Deployment, Hyperparameter Tuning, Apache Spark, MLflow, Pandas, NumPy

**Cloud & Tools:** Microsoft Azure (AI Engineer Associate - Certified), AWS S3, Google BigQuery, Jupyter, VS Code, Tableau, Power BI, Technical Documentation, Flowcharts & System Diagrams

**Testing & QA:** Pytest, Pylint, Unit Testing, Test Automation, Shell Scripting, Linux

**Tools & Collaboration:** Jira, Confluence, Bugzilla, Postman, GitHub, MySQL Workbench, PuTTY

## EXPERIENCE

**GenAI Application Developer - Graduate Practicum** | University of North Texas | Denton, TX July 2025 - Dec 2025

- Designed and developed a GenAI knowledge assistant using Python, FastAPI, LangChain, FAISS, and Hugging Face to enable document-based question answering through Retrieval-Augmented Generation.
- Built an end-to-end RAG pipeline for document ingestion, text preprocessing, chunking, embedding generation, vector indexing, similarity search, context retrieval, and structured response generation.
- Developed REST API endpoints using FastAPI to support document upload, user queries, retrieval workflows, and JSON-based responses for downstream application integration.
- Implemented prompt engineering techniques including role-based prompts, few-shot examples, structured output templates, and response validation rules to improve answer consistency and reduce unsupported responses.
- Integrated MLflow to track experiments, compare prompt versions, evaluate retrieval quality, monitor model behavior, and document results across multiple testing scenarios.
- Created validation test cases to evaluate retrieval accuracy, response relevance, edge cases, API reliability, and output formatting before final project delivery.
- Documented system architecture, API specifications, setup instructions, data flow diagrams, model evaluation notes, and project limitations to support production-style handoff.
- Applied software engineering best practices including modular code structure, Git-based version control, reusable components, testing workflows, and clear technical documentation.

*Environment: Python, FastAPI, LangChain, FAISS, Hugging Face, Azure AI, MLflow, REST APIs, Prompt Engineering, JSON, Git, Vector Search*

**Python Developer - Backup & Recovery Automation** | Client: Ericsson | Hyderabad, India Jan 2022 - May 2024

- Developed and maintained Python-based automation scripts to support enterprise backup and recovery operations using Symantec NetBackup, reducing manual effort across installation, configuration, and validation workflows.
- Automated NetBackup software installation, storage unit creation, backup policy configuration, and server recovery preparation by converting repetitive operational tasks into reusable Python scripts.
- Ported legacy shell scripts into modular Python scripts, improving maintainability, readability, error handling, and long-term supportability for backup automation processes.

- Built and enhanced test automation frameworks using Python and Pytest to validate backup installation steps, policy creation, storage configuration, and recovery-related workflows.
- Wrote unit tests, test scenarios, and validation scripts to verify automation reliability, detect configuration issues, and support continuous testing across sprint deliverables.
- Participated in Python design discussions, technical reviews, and peer code review sessions to improve script quality, performance, and coding standards.
- Collaborated with stakeholders, client teams, and internal project members to understand backup and recovery requirements, clarify technical expectations, and deliver sprint-based enhancements.
- Supported Agile delivery by creating and tracking user stories, updating task status in Jira, documenting progress, and contributing to sprint planning and review meetings.
- Prepared CPI and technical documentation, including setup steps, automation workflows, test results, and implementation notes for client reference and knowledge transfer.
- Conducted client demos to present completed automation features, explain workflow improvements, and gather feedback for future enhancements.

*Environment: Python, Shell Scripting, Pytest, Pylint, Flask, Django, MySQL, REST APIs, Postman, GitHub, Linux, PuTTY, Jira, Confluence, Bugzilla, Agile, MySQL Workbench*

## PROJECTS

**Intelligent Content Classification System - Full SDLC** | *Scikit-learn* | *CNN* | *SVM* | *NLP* | *Flask* | *MLflow* | *Power BI* | *Git*

- Led end-to-end software development - from requirements analysis and system design through coding, testing, debugging, and deployment - delivering a multi-algorithm NLP classifier achieving 95%+ accuracy on large-scale datasets.
- Developed automated test cases and validation scenarios; ensured error-free code through systematic debugging, code reviews, and comprehensive documentation of specifications, activities, and status reports.
- Deployed via Flask REST API with clean, modular, well-commented Python code following OOP principles; built Power BI dashboards to communicate system performance metrics to non-technical stakeholders.

**Multimodal Deep Learning Application - Speech Analysis** | *PyTorch* | *CNN* | *RNN/LSTM* | *MFCC* | *Streamlit* | *Data Augmentation* | *Git*

- Designed, developed, and deployed a real-time speech emotion recognition application - architecting the CNN-RNN system, engineering acoustic features (MFCC, mel-spectrogram), and optimizing performance through systematic hyperparameter tuning.
- Achieved 85%+ accuracy on 10,000+ labeled samples; prepared detailed technical documentation including architecture diagrams, code comments, and performance reports; deployed as a live Streamlit web application.

## CERTIFICATIONS

**Microsoft Certified: Azure AI Engineer Associate** | Microsoft — Issued 2024

**Generative AI with Large Language Models** | DeepLearning.AI / Coursera — Issued 2024